WORKSHEET

Chapter 9: Sample proportion calculations

1 Find the mean, variance and standard deviations of sample proportions for samples with the following probabilities and number in each sample.

a $p = 0.3$ and $n = 40$

b $p = 0.8$ and $n = 15$

c $p = 0.72$ and $n = 130$

d $p = 0.35$ and $n = 68$

2 The proportion of people between the ages of 18 and 22 who are full or part-time students is $\frac{2}{3}$.

Random samples of 16 people between the ages of 18 and 22 are selected. What is the mean and standard deviation of the sample proportion?

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3 It is known that 20% of all televisions sold at all TV World stores across Australia are sold with extended warranty. Samples of 25 people who recently bought televisions from TV world were taken and the proportion of those who took up the extended warranty was calculated. What is the expected proportion and standard deviation of the sample proportion?

4 A card is selected from a standard deck of playing cards and then replaced. Samples of 40 such selections are performed and the proportion of times the card is a face card (J, Q or K) is calculated for each sample. What is the mean and standard deviation of the sample proportions for selecting a face card?

5 Approximately 15% of the Australian population is known to be left-handed. Samples of 100 people from across Australia are surveyed to determine the proportion who are left-handed. What is the expected proportion and standard deviation of the sample proportion?

6 It is known that 35% of all Australian university students applied for entrance to universities other than the one they are attending. Samples of 1500 university students from across Australia are surveyed to determine the proportion who applied to other universities other than the one they are attending. What is the expected proportion and standard deviation of the sample proportion?

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Answers

1 **a** Mean $= 0.3$, variance $\approx 0.005\ 25$, standard deviation $\approx 0.072\ 46$

b Mean $= 0.8$, variance $\approx 0.010\ 67$, standard deviation ≈ 0.1033

c Mean $= 0.3$, variance $\approx 0.001\ 551$, standard deviation $\approx 0.039\ 38$

d Mean $= 0.3$, variance $\approx 0.003\ 346$, standard deviation $\approx 0.057\ 84$

2 Mean $= \frac{2}{3}$, standard deviation ≈ 0.1179

3 Expected proportion $= 0.2$, standard deviation $= 0.08$

4 Mean $= \frac{3}{13}$, standard deviation $\approx 0.066\ 62$

5 Expected proportion $= 0.2$, standard deviation $\approx 0.035\ 71$

6 Expected proportion $= 0.35$, standard deviation $\approx 0.012\ 32$

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